

Executive Summary

This is both an updated study of clinical drug development success rates from our 2016 report, and an expansion into the drivers of success with the addition of machine learning modeling to analyze the predictive factors contributing to drug development. A total of 12,728 clinical and regulatory phase transitions were recorded and analyzed from 9,704 development programs over the last decade (2011–2020), across 1,779 companies in the Biomedtracker database. Phase transitions occur when a drug candidate advances into the next phase of development or is suspended by the sponsor. By calculating the number of programs progressing to the next phase vs the total number progressing and suspended, we assessed the success rate at each of the four phases of development: Phase I, II, III, and regulatory filing. Having phase-by-phase data in hand, we then compared groups of diseases, drug modalities, and other attributes to generate the most comprehensive analysis yet of biopharmaceutical R&D success.

This work was made possible due to the years of clinical program monitoring and data entry by Informa Pharma Intelligence's Biomedtracker, which subsequently populates the purpose-built Probability of Technical Success (PTS) tool, Pharmapremia. BIO has long worked with Biomedtracker to calculate success rates based on this data. More recently, BIO and Biomedtracker teamed up with QLS Advisors, a team of expert data scientists who apply machine learning (ML) and artificial intelligence (AI) to clinical trials data and drug properties to assess the features that contribute the most amount, positively or negatively, to the probability of approval. The computational analyses of over 200 different drug, trial, indication, and sponsor metrics that influence clinical success rates were applied in a predictive capacity to produce risk assessments and evaluations of R&D assets with increased accuracy.

Key Takeaways

- The overall likelihood of approval (LOA) from Phase I for all developmental candidates over 2011–2020 was 7.9%.
- Phase II development remains the largest hurdle in drug development, with just 28.9% of candidates achieving this critical phase transition.
- Of the 14 major disease areas, Hematology therapies had the highest LOA from Phase I (23.9%), representing a seven-fold increase over the least successful group, Urology (3.6%).
- Immuno-oncology therapies provide a rare pocket of success in oncology R&D with an overall LOA of 12.4% vs 5.3% for all oncology approaches.
- Rare disease therapies were notably successful with an overall LOA of 17.0%.
- Chronic, high prevalence disease therapies were less successful with an overall LOA of 5.9%.
- Biological complexity in drug modalities generally leads to higher LOA, with CAR-T and RNA interference achieving the highest LOAs of 17.3% and 13.5%, respectively.
- Development programs with trials employing patient preselection biomarkers have two-fold higher LOAs (15.9%), driven by a Phase II success rate of nearly one-in-two.
- The top contributing factors toward phase success are disease indication, target, modality, and drug novelty.
- On average, it takes 10.5 years for a Phase I asset to progress to regulatory approval. Disease areas with above-average LOAs tend to have the shortest development timelines.

Part 1. Phase Transition Success and Likelihood of Approvals

Success rates for individual phases of the drug development process were determined by dividing the number that successfully advanced to the next phase by the total number advanced and suspended. This “advanced and suspended” number is often referred to as “n” in this report and should be taken into account when drawing conclusions from the success rate results.

Consistent with previous studies of drug development phase transition success rates, we found Phase II success rates to be far lower than any other phase.^{1,2} Phase I and III rates were substantially higher than Phase II, with Phase I slightly lower than Phase III. The highest success rate of the four development phases was the New Drug Application (NDA)/ Biologic License Application (BLA) filing phase (**Figure 1**).

The Phase I transition success rate was 52.0% (n=4,414). As this phase is typically conducted for safety testing and is not dependent on efficacy for candidates to advance, it is common for this phase to have a higher success rate among the clinical phases across most categories analyzed in this report. However, Phase I success rates also may benefit from delayed reporting or omission bias, as some larger companies may not deem failed Phase I programs as material and thereby not report them in the public domain.

The Phase II transition success rate (28.9%, n=4,933) was substantially lower than Phase I, and the lowest of the four phases studied. As this is generally the first stage where proof-of-concept is deliberately tested in human subjects, Phase II consistently has the lowest success rate of all phases. This is also the point in development where industry must decide whether to pursue large, expensive Phase III studies, or to terminate development – this may be done for multiple reasons, including commercial viability. The second-highest phase transition success rate was found in Phase III (57.8%, n=1,928). This is significant as most company-sponsored Phase III trials are the longest and most expensive trials to conduct.

The probability of FDA approval after submitting an NDA or BLA, taking into account re-submissions, was 90.6% (n=1,453).

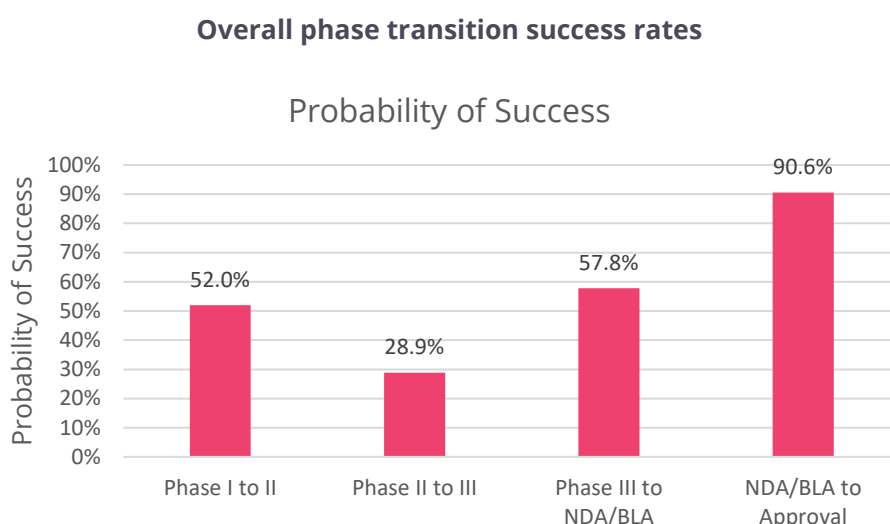


Figure 1: Phase transition success rates from Phase I for all diseases, all modalities. Source: Biomedtracker® and Pharmapremia®, 2020.

¹ Hay M, Thomas D, Craighead JL, Economides C, Rosenthal J (2014). Clinical development success rates for investigational drugs. *Nature Biotechnology*, 32(1), 40–51. doi: 10.1038/nbt.2786

² Thomas D, Burns J, Audette J, Carroll A, Dow-Hygelund C, Hay M (2016) Clinical Development Success Rates 2006-2015. Available [here](#) [Accessed 15 January 2021].

Phase Transition Success – By Disease Area

Major disease areas were segmented according to the convention used by Biomedtracker and categorized into 21 major diseases and 623 indications for the 2011–2020 timeframe. For reporting at the disease area level, in **Figure 2** we analyzed 14 major groupings: Allergy, Autoimmune, Cardiovascular, Endocrine, Gastroenterology(non-IBD), Hematology, Infectious disease, Metabolic, Neurology, Oncology, Ophthalmology, Psychiatry, Respiratory, and Urology. The remaining disease areas were placed into the “Other” category. This includes Dermatology, Renal, Obstetrics, Rheumatology (for non-autoimmune indications), ENT/Dental, and Orthopedics. Beneath these major disease areas are 573 indications, which will be analyzed and discussed in subsequent reports.

Phase transition success rates by disease area

Phase Success	Phase I to II		Phase II to III		Phase III to NDA/BLA		NDA/BLA to Approval	
	n	Phase POS	n	Phase POS	n	Phase POS	n	Phase POS
Hematology	92	69.6%	106	48.1%	82	76.8%	72	93.1%
Metabolic	136	61.8%	149	45.0%	66	63.6%	48	87.5%
Infectious disease	403	57.8%	414	38.4%	197	64.0%	156	92.9%
Others	154	63.6%	228	38.6%	90	60.0%	69	88.4%
Ophthalmology	88	71.6%	200	35.5%	82	51.2%	45	91.1%
Autoimmune	413	55.2%	471	31.4%	219	65.3%	202	94.1%
Allergy	55	56.4%	92	28.3%	34	64.7%	20	100.0%
Gastroenterology	45	46.7%	73	34.2%	35	57.1%	33	90.9%
All indications	4414	52.0%	4933	28.9%	1928	57.8%	1453	90.6%
Respiratory	179	55.9%	215	21.9%	62	64.5%	45	95.6%
Psychiatry	150	52.7%	164	26.8%	71	56.3%	57	91.2%
Endocrine	319	43.3%	293	26.6%	151	66.2%	124	86.3%
Neurology	516	47.7%	504	26.8%	226	53.1%	165	86.7%
Oncology	1628	48.8%	1732	24.6%	495	47.7%	324	92.0%
Cardiovascular	214	50.0%	252	21.0%	105	55.2%	80	82.5%
Urology	22	40.9%	40	15.0%	13	69.2%	13	84.6%

Figure 2: Phase transition success rates by disease area. The n value is the total ‘Advanced or Suspended’ transitions of all phases used to calculate LOA. ‘POS’ is the probability of successfully advancing to the next phase. The ordering of disease areas is consistent with the overall likelihood of approval from Phase I, which is analyzed later in Figure 5. Source: Biomedtracker® and Pharmapremia®, 2020

Phase I Transition Success Rates

Success rates by disease area for Phase I ranged from 40.9% to 71.6%, with the average for all disease indications coming in at 52.0%. Ophthalmology and Hematology were both well above the average rate, achieving respective successful Phase I transitions of 71.6% (n=88) and 69.6% (n=92). With the exception of these two major disease areas, the remainder were all within a reasonable distance from the mean.

Phase II Transition Success Rates

In every disease area, Phase II had the lowest transition success rate of the four phases. As shown in **Figure**

3, Phase II success rates ranged from a high of 48.1% (Hematology, n=106) to a low of 15.0% (Urology, n=40). This 33% range of disparity between major disease areas at the Phase II transition is the major contributor to the observed divergences in overall likelihood of approval (LOA), as discussed in the next section. With only Hematology and Metabolic (45.0%) coming close to achieving a one-in-two success rate, these two disease areas are also the leaders when calculating the overall LOA from Phase I, as shown later in **Figure 5**.

Similar to Phase I transition success, the lowest-performing disease groups were Urology, Cardiovascular (21.0%), and Oncology (24.6%). Relative to its placement for Phase I success rate, the Gastroenterology category performed better in Phase II (34.2%), above the average for all indications.

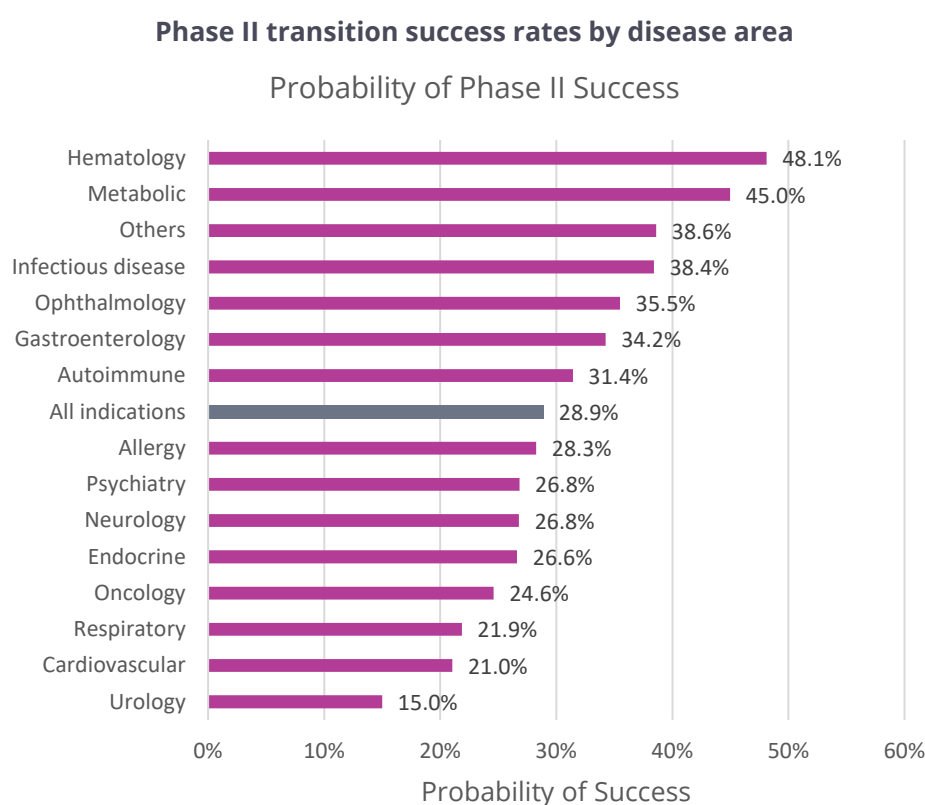


Figure 3: Phase II transition success rates by disease area. Source: Biomedtracker® and Pharmapremia®, 2020

Phase III Transition Success Rates

For Phase III transition success rates, Oncology had the lowest transition success rate (47.5%, n=495). As seen in Figure 4, the Phase III success rates for the remaining 13 specific disease areas were each above 50%.

Five other disease groups besides Oncology ranked below the average Phase III transition success rate of 57.8%, including Ophthalmology (51.2%, n=82) which in the earlier phases performed better than the average across all diseases. Conversely, Urology, which had the lowest transition rates in Phase I and II, had the second-highest Phase III success rate behind Hematology at 69.2%, albeit with a very low n of 13 transitions. All of the disease groups with below-average Phase III success rates had disease indications with large patient populations. Later in this report, we will analyze these high prevalence diseases and compare their success rates to those of therapies for rare diseases.

Oncology sub-category phase transition success rates and LOA

Phase Success	Phase I to II		Phase II to III		Phase III to NDA/BLA		NDA/BLA to Approval	
	POS n	Phase POS	POS n	Phase POS	POS n	Phase POS	POS n	Phase POS
Hematologic	425	50.1%	449	27.8%	120	60.0%	100	90.0%
Solid	1145	48.9%	1261	23.4%	364	42.9%	212	92.9%
IO	275	64.0%	244	40.2%	98	49.0%	62	98.4%

Likelihood of Approval	Phase I to Approval		Phase II to Approval		Phase III to Approval		NDA/BLA to Approval	
	LOA n	Phase LOA	LOA n	Phase LOA	LOA n	Phase LOA	LOA n	Phase LOA
Hematologic	1094	7.5%	669	15.0%	220	54.0%	100	90.0%
Solid	2982	4.6%	1837	9.3%	576	39.8%	212	92.9%
IO	679	12.4%	404	19.4%	160	48.2%	62	98.4%

Figure 7: Oncology sub-category phase transition success rates and LOA. The n value is the total 'Advanced or Suspended' transitions of all phases used to calculate LOA. 'POS' is the probability of successfully advancing to the next phase, whereas 'Phase LOA' is the probability of FDA approval for drugs from this phase of development. Note: Hematologic and solid tumor types are mutually exclusive categories, while the IO drug class includes transitions from across both categories. IO drugs for solid tumors had 3x the transitions as IO hematological cancers but have similar success rates. Source: Biomedtracker® and Pharmapremia®, 2020

Rare and Chronic High Prevalence Diseases

Within the Biomedtracker database, rare diseases can be identified based on meeting either or both of the following standard criteria: affecting fewer than 200,000 people in the US, or prevalence of 1 in 2,000 people in the EU. As 43% of the Oncology transitions are for rare indications, all Oncology indications were removed to make this rare disease analysis more concentrated on inborn genetic disorders. For chronic diseases, we first obtained a list of conditions from the CMS Chronic Conditions Data Warehouse (CCW). We removed any cancer indications, then identified those diseases with greater than 1 million patients affected in the United States.

Rare disease vs. highly prevalent chronic disease success rates

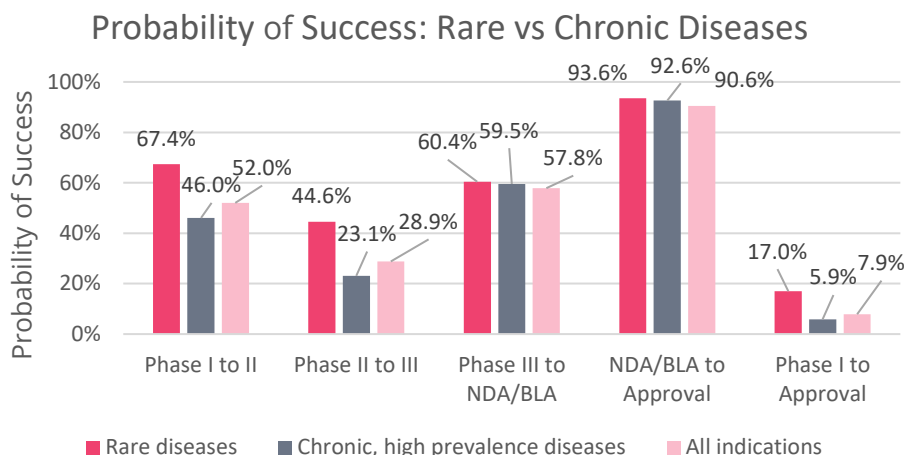


Figure 8a: Non-oncology rare disease and highly prevalent chronic disease phase transition success rates and LOA. Chart of phase transition success rates and LOA from Phase I. Source: Biomedtracker® and Pharmapremia®, 2020

Phase Success	Phase I to II		Phase II to III		Phase III to NDA/BLA		NDA/BLA to Approval	
	POS n	Phase POS	POS n	Phase POS	POS n	Phase POS	POS n	Phase POS
CAR-T	43	44.2%	17	58.8%	3	66.7%	4	100.0%
siRNA/RNAi	40	70.0%	38	28.9%	6	66.7%	3	100.0%
Monoclonal antibody	804	54.7%	740	34.1%	310	68.1%	282	95.4%
ADCs	103	41.7%	53	41.5%	16	62.5%	12	100.0%
Gene therapy	27	51.9%	57	38.6%	10	50.0%	2	100.0%
Vaccine	129	52.7%	117	31.6%	43	58.1%	27	100.0%
Protein	246	51.6%	288	33.0%	149	61.7%	117	89.7%
Peptide	234	53.0%	218	28.4%	100	60.0%	67	88.1%
Small molecule	2308	52.6%	2896	28.0%	1118	56.9%	849	89.5%
Antisense	69	60.9%	70	20.0%	14	64.3%	9	66.7%

Likelihood of Approval	Phase I to Approval		Phase II to Approval		Phase III to Approval		NDA/BLA to Approval	
	LOA n	Phase LOA	LOA n	Phase LOA	LOA n	Phase LOA	LOA n	Phase LOA
CAR-T	67	17.3%	24	39.2%	7	66.7%	4	100.0%
siRNA/RNAi	87	13.5%	47	19.3%	9	66.7%	3	100.0%
Monoclonal antibody	2136	12.1%	1332	22.1%	592	64.9%	282	95.4%
ADCs	184	10.8%	81	25.9%	28	62.5%	12	100.0%
Gene therapy	96	10.0%	69	19.3%	12	50.0%	2	100.0%
Vaccine	316	9.7%	187	18.4%	70	58.1%	27	100.0%
Protein	800	9.4%	554	18.3%	266	55.4%	117	89.7%
Peptide	619	8.0%	385	15.0%	167	52.8%	67	88.1%
Small molecule	7171	7.5%	4863	14.3%	1967	50.9%	849	89.5%
Antisense	162	5.2%	93	8.6%	23	42.9%	9	66.7%

Figure 10b: LOA from Phase I for drugs based on modality. Table of phase transition success and likelihood of approval by modality with corresponding n values. The n value is the total 'Advanced or Suspended' transitions of all phases used to calculate LOA. 'POS' is the probability of successfully advancing to the next phase, whereas 'Phase LOA' is the probability of FDA approval for drugs from this phase of development. *n values for vaccine differs slightly from Figure 9 owing to variable classification of cellular vaccines. Source: Biomedtracker® and Pharmapremia®, 2020

Patient Preselection Biomarkers

A greater understanding of human disease – whether at the molecular or genomic level – ultimately leads to the investigation of personalized medicine. Indications are increasingly segmented by biomarkers in order to match patients with the treatments most likely to show the greatest benefit, according to the underlying drug mechanism and disease pathophysiology. We identified 767 phase transitions out of 12,728 (6%) that incorporated patient preselection biomarkers in their corresponding clinical trial design. This was accomplished by mapping Informa Pharma Intelligence's Biomedtracker and Trialtruve databases, to provide the supplemental level of clinical trial detail. For a detailed description please refer to the Methods section.

The LOA from Phase I segmented by biomarkers can be found in **Figure 11**. Drug development programs with trials employing patient preselection biomarkers have a two-fold higher LOA (15.9%) than those that do not (7.6%).